

Layer Separation in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

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April 14, 2026 — Version 1

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Abstract

This document defines the separation of structural layers within Dimensional Human Field Theory (DHFT). The system is specified over invariant variables Load (L), Capacity (C), and Boundary (B).

The relations defined here establish distinctions between system structure, formal relations, measurement, and application. Layer separation is treated as a structural constraint of the system.

I. Scope

This document defines the structural separation between layers of description within DHFT. It specifies the conditions under which system definition, formalization, measurement, and application remain distinct.

No interpretive or domain-specific constructs are introduced.

II. System Layer

The system layer defines the minimal structure of DHFT: Load, Capacity, and Boundary.

III. Formal Layer

The formal layer defines relations governing system behavior, including stability conditions, boundary-regulated flow, and temporal accumulation. It does not alter the invariant variable set.

IV. Structural Extension Layer

This layer elaborates internal structure, including boundary conditions and identity, without altering variables or system definition.

V. Measurement Layer

The measurement layer defines derived constructs for evaluating system performance.

NSI is defined within this layer.

Measurement constructs introduce no variables and do not modify system structure.

VI. Application Layer

The application layer consists of domain-specific implementations.

Application varies across context but remains reducible to the minimal system.

VII. Non-Equivalence Across Layers

No layer is reducible to another.

Layer collapse produces a non-admissible system description.

VIII. Constraint

All valid uses of DHFT preserve separation between layers as defined in this document. Any description that conflates system, structure, measurement, or application violates admissibility conditions.

IX. System Reference

The system defined in this document is specified in *Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System*, DOI: 10.5281/zenodo.19075948.

All valid system descriptions reduce to Load, Capacity, and Boundary.

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical series.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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